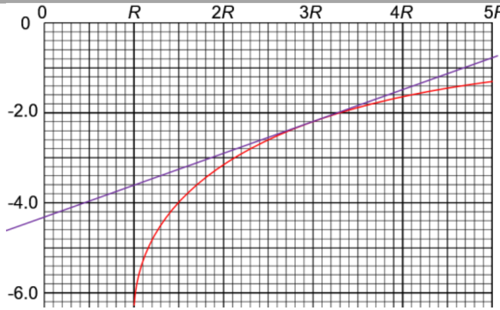
$E_k = 0.5\ K$ at minimum point clearly shown and labelled
$E_k = K$ at S and F clearly shown and labelled

3(a) work done per unit mass in moving small test mass from infinity to the point in the gravitational field

Qns	
3(b)	

$2R$ from earth surface → $x = 3R$

Method 1:
gradient at $x = 3R$:

$$\frac{y_1 - y_2}{x_1 - x_2} = \frac{\left[(-0.8)-(-4.3)\right]\times 10^7}{(5-0)\left(6.4\times 10^6\right)} = 1.09 \text{ N kg}^{-1}$$

$$|F| = |m_{\text{sat}}g| = m_{\text{sat}}\left|\frac{d\phi}{dr}\right| = (430)\left(1.09 \text{ N kg}^{-1}\right) = 470 \text{ N}$$

Method 2:

$$\phi_{\text{surface}} = -6.3\times 10^7 \text{ J kg}^{-1} = -\frac{GM_E}{R}$$

$$M_E = -\frac{R\phi_{\text{surface}}}{G} = -\frac{\left(6.4\times 10^6\right)\left(-6.3\times 10^7\right)}{6.67\times 10^{-11}}$$
$$= 6.0\times 10^{24} \text{ kg}$$

$$|F| = G\frac{M_E m_{\text{sat}}}{(3R)^2}$$
$$= \left(6.67\times 10^{-11}\right)\frac{\left(6.0\times 10^{24}\right)(430)}{\left(3\left(6.4\times 10^6\right)\right)^2}$$
$$= 470 \text{ N}$$

Qns	
3(c)(i)	Method 1:

$$\phi_{\text{surface}} = -6.3\times10^{7}\ \text{J kg}^{-1} = -\frac{GM_E}{R}$$

$$M_E = -\frac{R\phi_{\text{surface}}}{G} = -\frac{\left(6.4\times10^{6}\right)\left(-6.3\times10^{7}\right)}{6.67\times10^{-11}}$$

$$= 6.0\times10^{24}\ \text{kg}$$

Method 2:

$$|F| = G\frac{M_E m_{\text{sat}}}{(3R)^2}$$

$$M_E = \frac{(F)(3R)^2}{Gm_{\text{sat}}}$$

$$= \frac{(470)\left(3\left(6.4\times10^{6}\right)\right)^2}{\left(6.67\times10^{-11}\right)(430)}$$

$$= 6.0\times10^{24}\ \text{kg}$$

3(c)(ii) vector sum of gravitational force by earth and by sun on satellite provides centripetal force

$$\frac{GM_{\text{Sun}}m_{\text{sat}}}{(R-y)^2} - \frac{GM_E m_{\text{sat}}}{y^2} = m_{\text{sat}}(R-y)\omega^2$$

$$G\left[\frac{M_{\text{Sun}}}{(R-y)^2} - \frac{M_E}{y^2}\right] = (R-y)\left(\frac{2\pi}{T}\right)^2$$

$$\left(6.67\times10^{-11}\right)\left[\frac{1.99\times10^{30}}{\left(1.50\times10^{11}-y\right)^2} - \frac{6.0\times10^{24}}{y^2}\right] =$$

$$(1.50\times10^{11}-y)\left(\frac{2\pi}{(365)(24)(60)(60)}\right)^2$$

$$y = 1.7\times10^{9}\ \text{m}$$

3(c)(iii) at mid point, $x = 7.5\times10^{10}$ m

$$\lg\left(-\phi_{\text{total}}\right) = 9.25$$

$$\phi_{\text{total}} = -10^{9.25}\ \text{J kg}^{-1}$$

$$\text{energy needed} = m\left(\phi_{\text{final}} - \phi_{\text{total}}\right)$$

$$= (1)\left(0-\left(-10^{9.25}\right)\right)$$

$$= 1.8\times10^{9}\ \text{J}$$

4(a) the *uniform magnetic flux density* which when acting *perpendicularly* to a straight